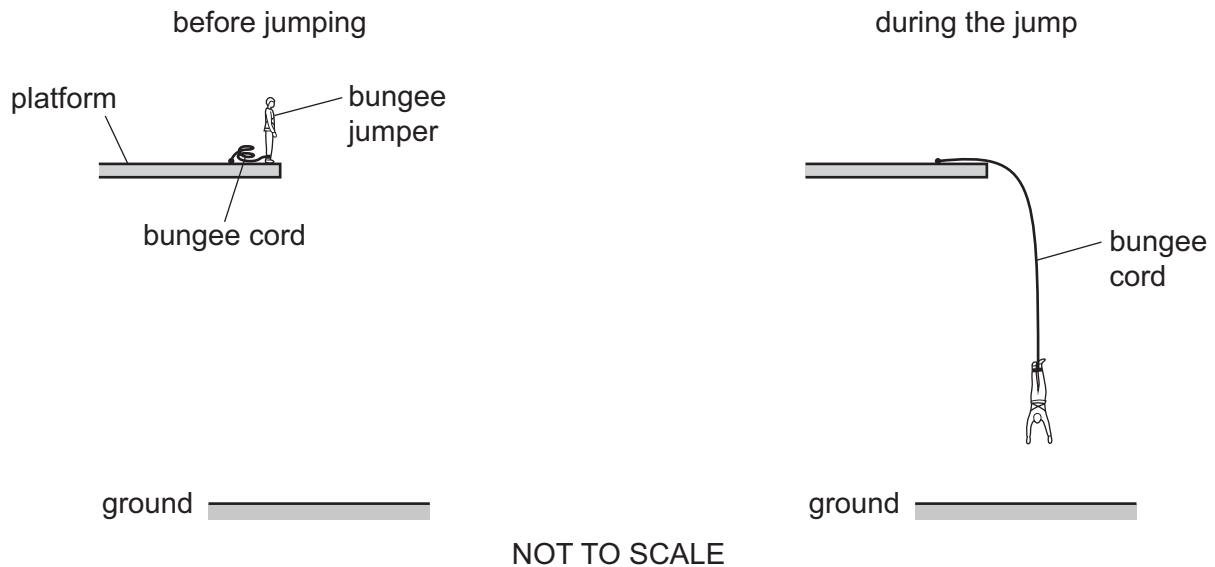


- 18** A bungee jumper jumps from a platform and is decelerated by an elastic bungee cord, as shown.



When the jumper makes the jump, his initial gravitational potential energy relative to the ground is converted into his kinetic energy and into elastic potential energy in the cord.

At which part of the jump are all three types of energy non-zero?

- A** on the platform before the jump
 - B** on the way down before the cord has started to extend
 - C** on the way down as he decelerates
 - D** at the bottom of the jump when he is stationary
- 19** A sailboat is pushed by the wind at a constant velocity v across a lake.

The force of the wind and the velocity of the sailboat are in the same direction.

Which additional information is required to determine the work done per unit time by the wind acting on the sailboat?

- A** the force exerted by the wind
- B** the distance travelled by the sailboat per unit time
- C** the total energy input to the sailboat by the wind
- D** the weight of the sailboat